

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

ROLL NO : 2520030255

NAME : V SAI NIKHIL

WEEK : 6

The screenshot displays the MySQL Workbench interface. The left sidebar contains the 'Navigator' pane with sections for MANAGEMENT (Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, Data Import/Restore), INSTANCE (Startup / Shutdown, Server Logs, Options File), and PERFORMANCE (Dashboard, Performance Reports, Performance Schema Setup). Below this are tabs for Administration, Schemas, and Information. The main editor area shows a SQL file with the following code:

```
90
91 DELIMITER $$
92 • CREATE TRIGGER prevent_salary_reduction
93 BEFORE UPDATE ON employees
94 FOR EACH ROW
95 BEGIN
96     IF NEW.salary < OLD.salary THEN
97         SIGNAL SQLSTATE '45000'
98         SET MESSAGE_TEXT = 'Salary reduction is not allowed'
99     END IF;
100 END $$
101 DELIMITER ;
102
103 • UPDATE employees
104 SET salary = 40000
105 WHERE emp_id = 101;
106
107
108 • UPDATE employees
109 SET salary = 60000
110 WHERE emp_id = 101;
111
112
```

The right sidebar shows a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help." Below this are tabs for Context Help and Snippets.

The bottom pane shows the 'Output' section with a table of execution results:

#	Time	Action	Message	Duration / Fetch
✓ 12	19:39:53	CREATE TRIGGER check_salary BEFORE INSE...	0 row(s) affected	0.031 sec
✗ 13	19:40:05	INSERT INTO employees VALUES (107, 'Ravi', 1...	Error Code: 1644. Salary cannot be less than 20000	0.016 sec
✓ 14	19:40:28	INSERT INTO employees VALUES (108, 'Meena'...	1 row(s) affected	0.015 sec
✓ 15	19:40:41	CREATE TRIGGER prevent_salary_reduction BE...	0 row(s) affected	0.015 sec
✓ 16	19:40:53	UPDATE employees SET salary = 40000 WHER...	0 row(s) affected Rows matched: 0 Changed: 0 ...	0.015 sec
✓ 17	19:41:12	UPDATE employees SET salary = 60000 WHER...	0 row(s) affected Rows matched: 0 Changed: 0 ...	0.000 sec

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas Information

No object selected

Query 1

```

30 END $$
31 DELIMITER ;
32
33
34 INSERT INTO employees
35 VALUES (100, 'Arjun', 1, 52000);
36
37
38 SELECT * FROM employee_audit;
39

```

Result Grid

emp_id	emp_name	action_type	action_time
1	100	Arjun	INSERT
2026-09-11 19:37:36			

Output

Action Output

Time	Action	Message	Duration / Rows
2 19:36:34	use hd		0 rows affected
3 19:36:37	CREATE TABLE employees(emp_id INT PRIMARY KEY, emp_name VARCHAR(50), dept_id INT, ...	0 rows affected	0.047 sec
4 19:36:51	CREATE TABLE employee_audit(audit_id INT AUTO_INCREMENT PRIMARY KEY, emp_id INT, em...	0 rows affected	0.018 sec
5 19:37:15	CREATE TRIGGER after_employee_insert AFTER INSERT ON employees FOR EACH ROW BEGIN INSE...	0 rows affected	0.015 sec
6 19:37:36	INSERT INTO employees VALUES (100, 'Arjun', 1, 52000)	1 rows affected	0.015 sec
7 19:37:49	SELECT * FROM employee_audit LIMIT 0, 1000	1 rows returned	0.000 sec / 0.000 sec

Object Info Session

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

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- Performance Schema Setup

Administration Schemas Information

No object selected

Query 1

```

32
33
34 INSERT INTO employees
35 VALUES (100, 'Arjun', 1, 52000);
36
37
38 SELECT * FROM employee_audit;
39
40 DELETE FROM employees
41 WHERE emp_id = 100;
42
43 DELIMITER $$
44 CREATE TRIGGER after_employee_delete
45 AFTER DELETE ON employees
46 FOR EACH ROW
47 BEGIN
48 INSERT INTO employee_audit
49 (emp_id, emp_name, action_type)
50 VALUES
51 (OLD.emp_id, OLD.emp_name, 'DELETE');
52
53 END $$
54 DELIMITER ;
55

```

Output

Action Output

Time	Action	Message	Duration / Rows
4 19:36:51	CREATE TABLE employee_audit(audit_id INT AUTO_INCREMENT PRIMARY KEY, emp_id INT, em...	0 rows affected	0.018 sec
5 19:37:15	CREATE TRIGGER after_employee_insert AFTER INSERT ON employees FOR EACH ROW BEGIN INSE...	0 rows affected	0.015 sec
6 19:37:36	INSERT INTO employees VALUES (100, 'Arjun', 1, 52000)	1 rows affected	0.015 sec
7 19:37:49	SELECT * FROM employee_audit LIMIT 0, 1000	1 rows returned	0.000 sec / 0.000 sec
8 19:38:20	DELETE FROM employees WHERE emp_id = 100	1 rows affected	0.000 sec
9 19:38:43	CREATE TRIGGER after_employee_delete AFTER DELETE ON employees FOR EACH ROW BEGIN INS...	0 rows affected	0.015 sec

Object Info Session

SQL Additions

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MySQL Workbench

Local instance MySQL80 x

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Administration Schemas

Information

Query 1

```
50 (OLD.emp_id, OLD.emp_name, 'DELETE');
51 END $$
52 DELIMITER ;
53
54
55 • UPDATE employees
56 SET salary = 60000
57 WHERE emp_id = 101;
58
59
60 DELIMITER $$
61 • CREATE TRIGGER after_employee_update
62 AFTER UPDATE ON employees
63 FOR EACH ROW
64 BEGIN
65 INSERT INTO employee_audit
66 (emp_id, emp_name, action_type)
67 VALUES
68 (NEW.emp_id, NEW.emp_name, 'UPDATE');
69 END $$
70 DELIMITER ;
71
72
```

SQL File 5* x

Limit to 1000 rows

SQLAdditions

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Context Help Snippets

No object selected

Object Info Session

Output

Action Output

#	Time	Action	Message	Duration / Fetch
6	19:37:36	INSERT INTO employees VALUES (106, 'Ajun', ...	1 row(s) affected	0.015 sec
7	19:37:49	SELECT * FROM employee_audit LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
8	19:38:20	DELETE FROM employees WHERE emp_id = 106	1 row(s) affected	0.000 sec
9	19:38:43	CREATE TRIGGER after_employee_delete AFTE...	0 row(s) affected	0.015 sec
10	19:39:11	UPDATE employees SET salary = 60000 WHER...	0 row(s) affected Rows matched: 0 Changed: 0 ...	0.000 sec
11	19:39:30	CREATE TRIGGER after_employee_update AFT...	0 row(s) affected	0.016 sec

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator: Query 1 procedures views SQL File 5* SQLAdditions

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Administration Schemas

Information

No object selected

Object Info Session

```

67 VALUES
68 (NEW.emp_id, NEW.emp_name, 'UPDATE');
69 END $$
70 DELIMITER ;
71
72
73 DELIMITER $$
74 CREATE TRIGGER check_salary
75 BEFORE INSERT ON employees
76 FOR EACH ROW
77 BEGIN
78 IF NEW.salary < 20000 THEN
79 SIGNAL SQLSTATE '45000'
80 SET MESSAGE_TEXT = 'Salary cannot be less than 20000';
81 END IF;
82 END $$
83 DELIMITER ;
84
85
86 INSERT INTO employees
87 VALUES (107, 'Ravi', 1, 15000);
88
89

```

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Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
8	19:38:20	DELETE FROM employees WHERE emp_id = 106	1 row(s) affected	0.000 sec
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11	19:39:30	CREATE TRIGGER after_employee_update AFT...	0 row(s) affected	0.016 sec
12	19:39:53	CREATE TRIGGER check_salary BEFORE INSE...	0 row(s) affected	0.031 sec
13	19:40:05	INSERT INTO employees VALUES (107, 'Ravi', 1...	Error Code: 1644. Salary cannot be less than 20000	0.016 sec

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Information

No object selected

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```

85
86 • INSERT INTO employees
87   VALUES (107, 'Ravi', 1, 15000);
88 • INSERT INTO employees VALUES (108, 'Meena', 1, 30000);
89
90
91 DELIMITER $$
92 • CREATE TRIGGER prevent_salary_reduction
93   BEFORE UPDATE ON employees
94   FOR EACH ROW
95   BEGIN
96     IF NEW.salary < OLD.salary THEN
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99     END IF;
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104   SET salary = 40000
105   WHERE emp_id = 101;
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107

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Context Help Snippets

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Action Output

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